

Robust Deepfake Detection - NTIRE 2026 Challenge Report

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Abstract—Deepfake detection has become increasingly important due to the rapid advancement of generative models and their misuse in spreading misinformation, identity theft, and digital fraud. However, many existing detection methods fail to generalize well under real-world conditions such as compression, noise, and unseen generation techniques. The NTIRE 2026 Robust Deepfake Detection Challenge focuses on addressing this gap by evaluating models on degraded and diverse data distributions. In this work, we explore a wide range of deep learning architectures for robust deepfake detection, including EfficientNet variants (B0, B2, B4, and EfficientNetV2), ConvNeXt models (tiny, small, and base), VGGFace2-based models, and hybrid architectures combining MobileNetV3, Swin Transformer, ResNet50d, and EfficientNet. Additionally, we investigate the impact of external data by training EfficientNet-B0 on an extended dataset of 40K images. Among all the approaches, ConvNeXt-based models demonstrate superior performance due to their ability to capture both local and global features effectively. Our results highlight the importance of model selection and data diversity in improving robustness for real-world deepfake detection.

I. INTRODUCTION

The rapid growth of generative artificial intelligence has led to the widespread creation of highly realistic synthetic media, commonly known as deepfakes. While these technologies have beneficial applications in entertainment and content creation, they also pose serious risks, including misinformation, impersonation, and security threats. As a result, developing reliable deepfake detection systems has become a critical research problem.

Traditional deepfake detection models often perform well on clean and controlled datasets but fail when exposed to real-world conditions such as image compression, noise, resizing, and unseen generative methods. To address these challenges, the NTIRE 2026 Robust Deepfake Detection Challenge provides a benchmark designed to evaluate the robustness and generalization capability of detection models under realistic degradations.

The challenge dataset consists of real and AI-generated images subjected to various transformations that simulate real-world scenarios. Participants are required to train models on the provided training data and submit predictions for validation and test sets through the official evaluation server. The performance is evaluated based on the model's ability to correctly distinguish between real and fake images under these challenging conditions.

In this work, we explore multiple deep learning architectures, including convolutional neural networks, transformer-based models, and hybrid approaches, to identify the most effective solution for robust deepfake detection. Our goal is

to analyze their performance and determine which model generalizes best to unseen and degraded data.

II. DATASET AND COMPETITION SPLITS

The NTIRE 2026 Robust Deepfake Detection Challenge provides a carefully designed dataset to evaluate not only detection accuracy but also the robustness and generalization ability of models. To prevent overfitting and ensure fair evaluation, the dataset is divided into multiple splits where the image sources and generative methods differ across each split.

A. Dataset Overview

The dataset consists of real and AI-generated images with varying levels of degradation. Unlike traditional datasets, this challenge introduces different data distributions across splits to simulate real-world conditions and unseen generation techniques.

Split	Samples	Labels	Degradations	Usage
Training	1,000	Yes	Simple	Model training
Validation	100	No	Additional	Daily evaluation
Public Test	1,000	No	Different	Final evaluation

TABLE I

SUMMARY OF DATASET SPLITS USED IN THE NTIRE 2026 CHALLENGE.

B. Training Data

The training set contains 1,000 samples consisting of both real and fake images. Each sample includes ground-truth labels, allowing supervised learning. The images in this split contain relatively simple degradations, enabling the model to learn the core differences between real and generated content.

C. Validation Data

The validation set consists of 100 images and is used for monitoring performance and tuning hyperparameters. Unlike the training set, this split includes additional degradations to simulate more challenging real-world scenarios. The ground-truth labels are not provided, and predictions must be submitted to the evaluation server. To maintain fairness, this data is strictly not used for training.

D. Public Test Data

The public test set contains 1,000 images and is used for final performance evaluation and leaderboard ranking. This split introduces different and less common degradations to assess how well the model generalizes to unseen conditions. The data is released only shortly before the submission deadline to prevent overfitting or manual tuning.

III. MODELS EVALUATED

In this work, we evaluated a diverse set of deep learning architectures to identify models that can generalize well under real-world degradations. The experiments include convolutional neural networks (CNNs), transformer-based models, hybrid architectures, and frequency-enhanced models. All models were initialized with pretrained weights and adapted for binary classification.

A. EfficientNet-based Models

EfficientNet architectures were used as baseline models due to their strong performance in image classification tasks. EfficientNet-B0 and EfficientNet-B2 were tested with modified classification heads for binary output.

Additionally, EfficientNet-B0 was trained using an external dataset of approximately 40,000 images to study the effect of additional data. However, this resulted in poor generalization (AUC: 0.40), likely due to distribution mismatch between the external dataset and the challenge dataset.

B. ConvNeXt Models

ConvNeXt architectures were extensively explored due to their ability to capture both local and global features effectively. Variants including ConvNeXt-Tiny and ConvNeXt-Base were tested with different classification heads and regularization strategies.

The best performance was achieved using ConvNeXt-Tiny with a simple and stable classification head consisting of Batch Normalization followed by a linear layer. This design reduced overfitting and achieved the highest validation AUC of 0.6916.

C. Face-Specific Model (VGGFace2)

A face recognition model based on InceptionResNetV1 pretrained on VGGFace2 was evaluated. The backbone was frozen, and only the classification head was trained (linear probing). Although this model captures strong facial identity features, it showed limited robustness to image degradations, achieving an AUC of 0.55.

D. Hybrid Architectures

To combine complementary feature representations, multiple hybrid models were explored. One approach combined MobileNetV3 (local feature extraction) with a Swin Transformer (global context modeling). Another approach fused ConvNeXt, EfficientNet, and ResNet representations.

These models aimed to improve robustness by leveraging both local texture information and global structural patterns. The MobileNetV3 + Swin Transformer hybrid achieved strong performance (AUC: 0.68), demonstrating the benefit of combining different architectural paradigms.

E. Frequency-based Models

Frequency-domain features were incorporated to capture artifacts introduced by generative models. Two main approaches were explored:

- FFT-based frequency branch combined with EfficientNetV2 backbone
- Edge-based frequency approximation using Sobel and Laplacian filters for stability

These models were designed to detect high-frequency inconsistencies common in synthetic images. While they improved robustness in some cases, they did not consistently outperform simpler ConvNeXt-based models.

F. ConvNeXt-Small with Frequency Fusion

To address the limitation of small training data (1,000 samples), a balanced architecture was explored using ConvNeXt-Small combined with a reduced frequency branch. Larger models such as EfficientNetV2-M were found to overfit due to their high parameter count.

The proposed model integrates:

- ConvNeXt-Small backbone for spatial feature extraction
- A compact FFT-based frequency branch to capture high-frequency artifacts
- A shallow fusion head with strong dropout regularization

The reduced frequency branch helped prevent overfitting while still capturing useful spectral features. This model achieved strong performance (AUC: 0.68) and demonstrated better stability compared to more complex hybrid models.

G. Performance Comparison

Table II summarizes the performance of all evaluated models on the validation set using AUC as the evaluation metric.

Model	Type	AUC
EfficientNet-B0	CNN	0.5912
EfficientNet-B2	CNN	0.57
EfficientNet-B0 (External Data)	CNN	0.40
ConvNeXt-Tiny (Dropout Head)	CNN	0.6852
ConvNeXt-Tiny (BatchNorm Head)	CNN	0.6916
ConvNeXt-Tiny (Variant)	CNN	0.65
ConvNeXt-Base (384)	CNN	0.63
VGGFace2 (InceptionResNetV1)	Face Model	0.55
EfficientNetV2 + FFT Branch	Hybrid	0.64
EfficientNet-B4 + Edge Branch	Hybrid	0.50
ConvNeXt-Tiny + Frequency Branch	Hybrid	0.66
MobileNetV3 + Swin Transformer	Hybrid	0.68
ConvNeXt-Small + Frequency Branch	Hybrid	0.68
ConvNeXt-Tiny (Augmented Variant)	CNN	0.60

TABLE II
COMPARISON OF DIFFERENT MODELS EVALUATED ON THE VALIDATION SET.

H. Public Test Evaluation

The top three models from the validation phase were evaluated on the public test set to assess generalization performance.

Model	Validation AUC	Test AUC
ConvNeXt-Tiny (BatchNorm Head)	0.6916	0.65
MobileNetV3 + Swin Transformer	0.68	0.5956
ConvNeXt-Small + Frequency Branch	0.68	0.686

TABLE III
COMPARISON OF VALIDATION AND TEST PERFORMANCE.

The results show a clear gap between validation and test performance, indicating overfitting in more complex models.

The ConvNeXt-Small with frequency branch achieved the best generalization and was selected as the final model.

IV. PROPOSED METHOD

A. Model Architecture

The final model is based on a dual-branch architecture that combines spatial and frequency-domain features for robust deepfake detection. The design captures both visual patterns and high-frequency artifacts present in synthetic images.

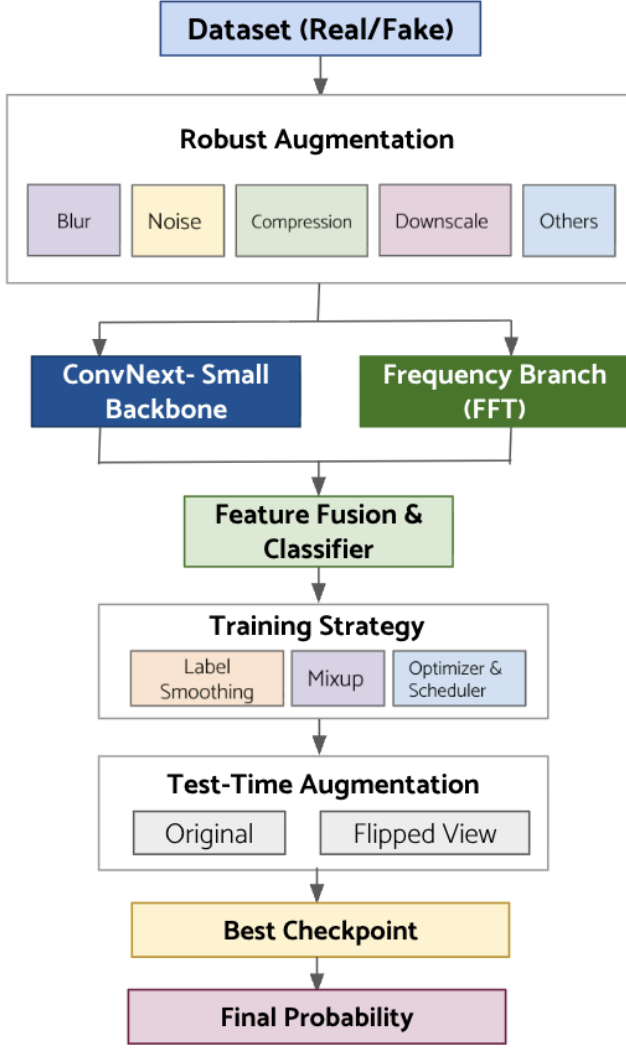


Fig. 1. Overview of the proposed dual-branch architecture combining ConvNeXt-Small and a frequency-based branch.

1) Architecture Overview:

2) *RGB Backbone*: The primary branch uses a ConvNeXt-Small backbone pretrained on ImageNet. The classification head is removed, and global average pooled features are extracted. This backbone provides strong spatial feature representations and is effective in handling degraded images.

3) *Frequency Branch*: A lightweight frequency branch is used to capture spectral artifacts introduced by generative models. The input image is converted to grayscale, followed by a 2D Fast Fourier Transform (FFT). The log-magnitude spectrum is computed and normalized before being processed through a small convolutional network.

This branch is intentionally kept compact to reduce overfitting while still capturing useful high-frequency information.

4) *Feature Fusion*: The outputs from both branches are concatenated and passed through a shallow fully connected head. The head includes Layer Normalization, Dropout, and GELU activation for improved generalization and stability.

B. Training Strategy

The training process was designed specifically for a small dataset:

- Backbone frozen for initial epochs, followed by fine-tuning
- Strong regularization using dropout and weight decay
- Mixup augmentation for improved generalization
- Label smoothing with binary cross-entropy loss
- Cosine annealing learning rate scheduler
- Test-time augmentation (TTA) during validation

Balanced sampling was applied using weighted sampling to ensure class balance.

C. Hyperparameters

Parameter	Value
Backbone	ConvNeXt-Small
Input Size	224×224
Batch Size	16
Epochs	30
Optimizer	AdamW
Learning Rate	1×10^{-4}
Weight Decay	5×10^{-2}
Loss Function	BCE with Label Smoothing
Mixup Alpha	0.2
Dropout	0.5
Scheduler	Cosine Annealing

TABLE IV

TRAINING HYPERPARAMETERS OF THE PROPOSED MODEL.

D. Inference Strategy

During inference, lightweight test-time augmentation (horizontal flip) was applied, and predictions were averaged. While ensemble methods were explored, the single best model provided more stable performance on the public test set.

E. Discussion

The proposed model achieved the best performance on the public test set (AUC: 0.686). The results show that combining spatial and frequency features, along with strong regularization, leads to better generalization compared to larger or more complex models.

V. RESULTS AND LEADERBOARD PERFORMANCE

A. Validation Leaderboard

44	bili0815	2026-02-25 15:49	563686	0.7384
45	jc555	2026-03-10 16:20	610425	0.7284
46	yang0201	2026-02-07 11:06	528482	0.7238
47	gwq_gwq_gwq	2026-02-11 11:32	536251	0.7212
48	dkhonker	2026-02-04 14:58	522557	0.7208
49	dallas_maverick	2026-03-04 12:26	589368	0.7204
50	juhyeonpark	2026-02-09 19:20	532391	0.7192
51	wenbin	2026-02-03 12:48	520328	0.7156
52	AI4Good	2026-03-16 10:19	628101	0.712
53	beautifulboy	2026-02-11 16:47	536754	0.7016
54	Ashwathi SNUC	2026-02-13 08:49	539729	0.6916
55	aksh_amirtharaj	2026-03-10 21:37	611382	0.6892
56	richhhh	2026-02-10 10:56	533997	0.6882
57	waxiangji	2026-02-10 09:11	533835	0.688
58	miketjc	2026-03-15 13:09	625088	0.6864
59	liuruqi	2026-02-15 08:43	542969	0.6616
60	lucamo	2026-03-16 04:39	627273	0.6616

Fig. 2. Public validation leaderboard results.

On the public validation set, the best-performing model achieved an AUC of 0.6916, securing a rank of **54 out of 92** participants. ConvNeXt-based models consistently outperformed other architectures, showing strong capability in handling degraded images.

B. Public Test Leaderboard

41	challenger004	2026-03-17 12:36	632553	0.8089
42	wxy666	2026-03-16 15:29	629167	0.8074
43	ag2617	2026-03-17 07:01	631618	0.8045
44	utktw1	2026-03-17 07:29	631695	0.8014
45	yuhdaddy	2026-03-17 13:10	632677	0.7995
46	arjulk	2026-03-18 20:08	636503	0.7958
47	ag16373	2026-03-17 05:31	631427	0.7883
48	utkars57609	2026-03-17 01:12	630839	0.7777
49	lucamo	2026-03-17 06:01	631490	0.7281
50	anasmall	2026-03-17 07:11	631645	0.7056
51	Aishwarya SNUC	2026-03-17 10:16	632183	0.686
52	yaoxin_jiang	2026-03-17 00:32	630777	0.6824
53	Ashwathi SNUC	2026-03-16 14:13	628876	0.6519
54	miketjc0316	2026-03-17 00:27	630765	0.6186
55	aksh_amirtharaj	2026-03-17 10:33	632224	0.5999
56	sharan	2026-03-17 10:21	632193	0.5956
57	akshara__161	2026-03-17 10:59	632285	0.4816

Fig. 3. Public test leaderboard results.

On the public test set, the proposed ConvNeXt-Small with frequency branch achieved an AUC of 0.686, resulting in a rank of **51 out of 57** participants. This model demonstrated better generalization compared to other approaches evaluated.

C. Discussion

A noticeable gap between validation and test performance was observed, indicating overfitting in several models. Simpler architectures with strong regularization and frequency-aware features showed better generalization to unseen data.

VI. CODE AVAILABILITY

The implementation of the proposed model, including training and inference pipelines, is publicly available on GitHub.

Repository:

https://github.com/aishwarya-41/deepfake_detection_ntire/tree/main/code_preliminary4_deepfake

The repository contains all necessary files to reproduce the results, including dataset preprocessing, model architecture, training scripts, and inference code.

VII. CONCLUSION

In this work, we explored multiple deep learning architectures for robust deepfake detection under real-world degradations as part of the NTIRE 2026 challenge. A wide range of models, including EfficientNet, ConvNeXt, hybrid architectures, and frequency-based approaches, were evaluated.

The results showed that model complexity alone does not guarantee better performance, especially with limited training data. Simpler architectures with strong regularization performed better in terms of generalization.

The proposed ConvNeXt-Small with frequency branch achieved the best performance on the public test set, demonstrating the effectiveness of combining spatial and frequency-domain features.

Future work can focus on improving robustness further using larger datasets, better augmentation strategies, and ensemble methods.